

CEHv13 AI Module 4- Enumeration

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Overview

Enumeration is the process of extracting detailed information about identified services, shares, and users on target systems. Unlike scanning, which identifies open ports and services, enumeration involves actively connecting to systems and performing queries to gather specific information that can be used for attacking a system.

-Enumeration is the extended form of scanning, which mainly focuses on port scanning and services running on those ports

-Enumeration is the process of extracting usernames, machine names, network resources, shares, and services from a system or network.

Objective

The objective of the lab is to extract information about the target organisation that includes, but is not limited to:

- Machine names, their OSes, services, and ports
- Network resources
- Usernames and user groups
- Lists of shares on individual hosts on the network
- Policies and passwords
- Routing tables
- Audit and service settings
- SNMP and FQDN details

Difference b/w Scanning and Enumeration -

Scanning focuses on identifying what exists: live hosts, open ports, running services, and basic configurations. It gives you a broad, surface-level picture of the environment with a moderate amount of interaction and a lower risk profile.

Whereas:-

Enumeration digs way deeper. Instead of just finding doors, it tries to open them and see what's inside. It pulls structured information like usernames, shares, domain details, SNMP data, DNS records, and other internal components.

Enumeration requires more protocol knowledge, generates far more noise, and is much easier for defenders to detect because it interacts aggressively with services.

Scanning answers “what’s there?” while enumeration answers “how does it work and what can I pull from it?”

Types of Enumeration:

1. NetBIOS Enumeration

- NetBIOS (Network Basic Input/Output System) allows applications to communicate over a local area network.
- Default Ports:
 - 137/UDP - NetBIOS Name Service
 - 138/UDP - NetBIOS Datagram Service
 - 139/TCP - NetBIOS Session Service
- Information Gathered:
 - Computer names and workgroup information
 - Shared resources and folders
 - User accounts and groups
 - System uptime and OS version
- Tools and Techniques:
 - nbtstat, enum4linux, nbtscan

2. SNMP Enumeration

- Simple Network Management Protocol used for network device management.
- Default Port: 161/UDP
 - 162/UDP - SNMP Trap
- Information Gathered:
 - System details, network devices, routing tables, running processes
- Tools:
 - snmpwalk, snmpenum

3. LDAP Enumeration

- Used for querying directory services like Active Directory.
- Information Gathered:
 - Usernames, group memberships, policies
- Tools:
 - ldapsearch, JXplorer

4. NTP Enumeration

- Network Time Protocol enumeration can reveal internal IPs, server time, and client information.
- Default Port: 123/UDP

- Tools:

- ntpq, ntpdc

5. SMTP Enumeration

- Used to identify valid user accounts on mail servers.

- Default Port: 25/TCP

- Commands and Tools:

- VRFY, EXPN commands

- smtp-user-enum, Telnet

6. DNS Enumeration

- Gathers subdomains, hostnames, and DNS records.

- Tools:

- nslookup, dig, dnsenum, fierce

7. RPC Enumeration

- Remote Procedure Call enumeration to discover Windows services, users, and groups.

- Default Port: 135/TCP

- Tools:

- rpcclient, enum4linux

- NetBIOS Enumeration

- Gathers network shares, user lists, and system names.

8. NTP Enumeration

- Discovers network time protocol configurations and connected hosts.

9. SMB Enumeration

- Lists shared files, printers, and services.

10. HTTP Enumeration

- Finds web directories, subdomains, and server details.

Enumeration Methodology:

- **Port Scanning**

Identify open ports and available services on the target.

- **Service Fingerprinting**

Determine versions and configurations of detected services.

- **Protocol-Specific Enumeration**

Use specialised tools and techniques for protocols like NetBIOS, SNMP, SMB, etc.

- **User and Group Enumeration**

Collect valid usernames and group information for potential password attacks.

- **Share Enumeration**

Identify accessible shared resources such as files and printers.

- **Banner Grabbing**

Retrieve detailed software information from service banners.

- **Data Analysis**

Consolidate, filter, and interpret gathered information for exploitation or defense planning.

Enumeration Tools and Techniques

1. NetBIOS enumeration with nmap

```
nmap -sU -p 137 --script nbstat target-ip
```

NetBIOS name lookup

```
nbtscan target-ip
```

Enumerate NetBIOS shares

```
smbclient -L //target-ip -N
```

Net command enumeration (Windows)

```
net view \\target-ip
```

```
net use \\target-ip\ipc$ "" /user:""
```

2. SNMP enumeration with snmpwalk

snmpwalk -c public -v1 target-ip

Enumerate system information

snmpwalk -c public -v1 target-ip 1.3.6.1.2.1.1

Enumerate network interfaces

snmpwalk -c public -v1 target-ip 1.3.6.1.2.1.2.2.1.2

Enumerate running processes

snmpwalk -c public -v1 target-ip 1.3.6.1.2.1.25.4.2.1.2

Enumerate installed software

snmpwalk -c public -v1 target-ip 1.3.6.1.2.1.25.6.3.1.2

SNMP brute force community strings

onesixtyone -c community.txt target-ip

SNMP enumeration script

snmp-check target-ip -c public

3. LDAP enumeration with ldapsearch

ldapsearch -x -h target-ip -p 389 -s base

Anonymous bind enumeration

ldapsearch -x -h target-ip -p 389 -s sub -b "dc=domain,dc=com"

Enumerate domain users

ldapsearch -x -h target-ip -p 389 -s sub -b "dc=domain,dc=com" "(objectclass=user)"

Enumerate domain groups

ldapsearch -x -h target-ip -p 389 -s sub -b "dc=domain,dc=com" "(objectclass=group)"

LDAP enumeration with nmap

nmap -p 389 --script ldap-rootdse target-ip

nmap -p 389 --script ldap-search target-ip

4. NTP enumeration with ntpq

`ntpq -c readlist target-ip`

`ntpq -c peers target-ip`

NTP version enumeration

`ntpdate -q target-ip`

NTP enumeration with nmap

`nmap -sU -p 123 --script ntp-info target-ip`

`nmap -sU -p 123 --script ntp-monlist target-ip`

5. Common SNMP enumeration commands:

- `snmpwalk -c public -v1 <target_ip>`
 - Performs a basic SNMP walk using community string "public" on SNMP version 1.
- `snmpwalk -c public -v2c <target_ip>`
 - SNMP walk using community string "public" on SNMP version 2c.
- `snmp-check -c public <target_ip>`
 - Retrieves system details and device information using the public community string.
- `snmpbulkwalk -c public -v2c <target_ip> <OID>`
 - Efficient bulk retrieval of SNMP information on specified OID range.
- `snmpget -c public -v1 <target_ip> <OID>`
 - Retrieves specific SNMP object value for the given OID.
- `snmpgetnext -c public -v1 <target_ip> <OID>`
 - Gets the next SNMP object value after the specified OID.

6. DNS Enumeration

DNS zone transfer

`dig axfr @target-ip domain.com`

`host -l domain.com target-ip`

DNS record enumeration

`dig any domain.com @target-ip`

`nslookup`

`> set type=any`

`> domain.com`

DNS brute force subdomain enumeration
dnsrecon -d domain.com -t brt
fierce -dns domain.com

DNS enumeration with nmap
nmap -p 53 --script dns-zone-transfer target-ip
nmap -p 53 --script dns-brute domain.com

7. RPC enumeration with rpcinfo

rpcinfo -p target-ip

RPC endpoint enumeration (Windows)
rpcinfo -T tcp target-ip

RPC enumeration with nmap
nmap -p 135 --script rpc-grind target-ip
nmap -p 111 --script rpc-grind target-ip

Windows RPC enumeration
rpcclient -U "" -N target-ip

8. Ftp Enumeration Techniques

FTP banner grabbing
telnet target-ip 21
nc target-ip 21

Anonymous FTP access
ftp target-ip
Username: anonymous
Password: (blank or email)

FTP enumeration with nmap
nmap -p 21 --script ftp-anon target-ip
nmap -p 21 --script ftp-bounce target-ip

FTP brute force

```
hydra -L users.txt -P passwords.txt target-ip ftp
```

9. SSH Enumeration

SSH banner grabbing
 telnet target-ip 22
 nc target-ip 22

SSH version enumeration
 ssh -V target-ip

SSH enumeration with nmap
 nmap -p 22 --script ssh-hostkey target-ip
 nmap -p 22 --script ssh-auth-methods target-ip

SSH user enumeration
 ssh-keyscan target-ip

Tools for Enumeration:

- Nmap – Service version detection.
- Enum4linux – SMB and NetBIOS enumeration.
- SMBclient – Access SMB shares.
- Net view – List network resources (Windows).
- SNMPwalk – SNMP data gathering.
- SMTP-user-enum – SMTP username enumeration.
- Dirbuster / Gobuster – Web directory enumeration.
- DNSenum / Fierce – DNS enumeration.
- Hping3 – Crafted packet generator and scanner, useful for stealth enumeration.

Bloodhound

Data collection with SharpHound
 .\SharpHound.exe -c all

Data collection with Python
 bloodhound-python -u username -p password -ns target-ip -d domain.com -c all

Start BloodHound
 neo4j console

bloodhound

LinEnum

Linux enumeration script

`chmod +x LinEnum.sh`

`./LinEnum.sh`

Thorough enumeration

`./LinEnum.sh -t`

Network Level Protection and Best Practices:

- Firewall Rules:

Block unnecessary ports and services to minimize attack surface.

- Network Segmentation:

Isolate critical systems to prevent lateral movement.

- VPN Access:

Require VPN for all administrative access.

- Port Security:

Disable unused network ports to prevent unauthorized connections.

Service Hardening:

- - Disable Unnecessary Services:
 - Remove or turn off services not required.
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- - Change Default Configurations:
 - Modify default settings and ports to reduce predictability.
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- - Strong Authentication:
 - Implement strong passwords and multi-factor authentication (MFA).
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- - Access Controls:
 - Limit user privileges and access on a need-to-know basis.

SNMP Security:

- Use SNMPv3:
Employ SNMPv3 with authentication and encryption.
- Custom Community Strings:
Avoid default community strings like "public" or "private".
- Access Control Lists (ACLs):
Restrict SNMP access by IP addresses.
- Disable SNMP:
Disable SNMP on devices where it is not needed.

SMB/NetBIOS Security:

- Disable SMBv1:
Use only SMBv2 or SMBv3, as SMBv1 is vulnerable.
- Null Session Protection:
Prevent anonymous connections.
- Share Permissions:
Properly configure SMB share and NTFS permissions.
- Network Authentication:
Require authentication for accessing shares.

LDAP Security:

- Bind Authentication:
Disable anonymous bind access.
- SSL/TLS Encryption:
Use LDAPS for encrypted LDAP communications.
- Access Controls:
Implement proper directory permissions.

Perform Enumeration using AI

Perform NetBIOS enumeration on target system

sgpt --shell "Perform NetBIOS enumeration on target IP 10.10.1.11"

Netbios enumeration

sgpt --shell "Get NetBIOS info for IP 10.10.1.11 and display the associated names"

sgpt --shell "Enumerate NetBIOS on target IP 10.10.1.22 with nmap"

SNMP enumeration

sgpt --chat enum --shell "Perform SNMP enumeration on target IP 10.10.1.22 using SnpWalk and display the result here"

sgpt --chat enum --shell "Perform SNMP enumeration on target IP 10.10.1.22 using nmap and display the result here"

Other Examples

- **sgpt --chat enum --shell** "Perform SNMP processes on target IP 10.10.1.22 using nmap and display the result here"
- **sgpt --chat enum --shell** "Perform SMTP enumeration on target IP 10.10.1.19."
- **sgpt --chat enum --shell** "Use Nmap to perform DNS Enumeration on target domain www.certifiedhacker.com"
- **sgpt --chat enum --shell** "Use dig command to perform DNS cache snooping on target domain www.certifiedhacker.com using recursive method. Use DNS server IP as 162.241.216.11"
- **sgpt --chat enum --shell** "Use dig command to perform DNS cache snooping on the target domain www.certifiedhacker.com using non-recursive method. Use DNS server IP as 162.241.216.11"
- **sgpt --shell** "Perform IPsec enumeration on target IP 10.10.1.22 with Nmap"
- **sgpt --shell** "Scan the target IP 10.10.1.22 for the port using SMB with Nmap"
- **sgpt --chat enum --shell** "Develop and execute a script which will automate various network enumeration tasks on target IP range 10.10.1.0/24"
- **sgpt --shell** "Use nmap script to perform ldap-brute-force on IP 10.10.1.22"
- **sgpt --shell** "Use Nmap to perform FTP Enumeration on www.certifiedhacker.com"